A Bayesian Model for Selective Social Learning in 12-year-old Children

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March 31st, 2026

Abstract

When children solve a problem, they are typically faced with multiple sources of information that may lead to the correct solution. But how do they know which sources to trust? Children cross-validate a source's information against their own understanding to assess whether that source's information is accurate. If they deem the source to be accurate, they will incorporate its information into their knowledge pool; otherwise, they will disregard it (Koenig, Clément, and Harris, 2004; Koenig and Woodward, 2010; Krogh-Jesperson and Echols, 2012). In developmental psychology, this mechanism is known as selective social learning. Critically, Bayesian models of cognition can capture this cross-validation mechanism that gives rise to selective social learning. Therefore, I wanted to examine whether a Bayesian model can replicate selective social learning in twelve-year-old children. To answer my question, I used a ball-and-urn task in which children estimated the proportion of white balls in their own urn. To do so, they could either draw one ball at a time from their personal urn, i.e., personal sampling, or consult another agent about the proportion of white balls in its urn, i.e., social sampling. To elicit selective social learning, I randomly assigned the children to one of two conditions: in the first condition, the other agent's urn was identical to the child's personal urn; in the second, the other agent's urn had more white balls than the personal urn. This task successfully elicited selective social learning in children as the number of personal samples in the first condition was much higher than the second condition. In addition, the Bayesian model obtained similar results, meaning that it approximated this selective social learning behaviour in children.

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Introduction

Suppose that you had to solve for $\mathbf{X}$ in the following problem:

$$\mathbf{X} = \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \otimes \begin{pmatrix} g & h \\ i & j \end{pmatrix}.$$

What would your instinct be? Would you immediately seek help from another person or would you try to solve it on your own? Now, suppose that you had to solve for $\mathbf{X}$ in the following problem: $\mathbf{X} = 5^2$. Does your instinct change? Importantly, each problem-solving method has its own costs and benefits. For example, solving a problem on your own may require a higher level of cognitive labour relative to asking another person for help. If the problem is too complex to solve on your own, you may seek help from another person. However, you may be more uncertain about whether their approach is correct. At the extreme, if you believe that this person is inaccurate, then you would disregard their solution. Interestingly, children also disregard information from inaccurate sources, which is called selective social learning. Koenig, Clément and Harris (2004) conducted a pivotal study that substantiated the existence of such behaviour in children. They presented three- and four-year-old children with an accurate labeler and an inaccurate labeler. Then, the children were presented with familiar objects, such as a ball, and, for each object, they observed how each labeler named the object. Then, the children were asked to name the object themselves. At the end of this familiarization stage, the experimenter asked each child to identify the accurate labeler and the inaccurate labeler. In the second stage, the children were exposed to objects that they have never seen before. For each object, the children followed the same procedure in the familiarization stage. The researchers found that the children who identified an informant as inaccurate across both stages did not name the novel objects by using the "inaccurate" agent's labels. This result suggests that children

tend to cross-validate the information provided by an agent with their own knowledge to determine whether this agent's information is accurate. Then, if they believe that the agent is (inaccurate)/(accurate), then they will (not incorporate)/(incorporate) the agent's labels into their knowledge pool. Krogh-Jesperson and Echols (2012) and Koenig and Woodward (2010) replicated these findings for similarly aged children. In addition, Koenig & Echols (2003) also found signs of selective social learning in 16-month-old infants; however, they adapted their experimental design to make it more comprehensible for infants. The infants repeatedly heard a human accurately or inaccurately label familiar objects. The researchers found that the infants looked significantly longer at the human when it inaccurately labeled an object, suggesting that the infants were surprised when the human mislabeled the object. This finding signals that infants cross-validate information given to them with their own understanding of the world very early on.

What are the mechanisms that drive selective social learning? Sobel and Kushnir (2013) suggest that children make inferences about the accuracy of social agents by determining whether a social agent's information agrees with their understanding of the world. To accomplish this goal, infants examine whether an agent's accuracy is consistent across many scenarios to infer its overall accuracy (Aslin & Newport, 2012; Denison & Xu, 2014). Critically, Bayesian models of cognition can express the mechanisms that give rise to selective social learning (Griffiths et al., 2006; Griffiths et al., 2024; Vul et al., 2014). Previous research has not explored whether Bayesian models can approximate selective social learning in twelve-year-old children. The reasoning behind this age range will be discussed later. Therefore, I am interested in determining whether selective social learning in twelve-year-old children can be approximated using a Bayesian model.

Concerning Bayesian models of cognition, prior research has found that human behavior in inference tasks, such as a ball-and-urn experiment, resembles that of an ideal Bayesian agent (Griffiths et al., 2006; Kolossa et al., 2015; Kolossa, 2016; Vul et al., 2014). This agent infers latent variables by combining prior knowledge with evidence from new

observed data (Griffiths et al., 2024). This process is expressed by Bayes' theorem:

$$P(h \mid d) \text{ is proportional to } P(d \mid h)\, P(h).$$

In essence, there exists a set of hypotheses $\mathcal{H}$ that can explain the observed data d. For each hypothesis h in $\mathcal{H}$, the Bayesian agent cares about how well the hypothesis h can explain the data d, denoted by $P(d \mid h)$, and how likely it would observe hypothesis h without observing any data, being $P(h)$. The latter is called a prior belief. These components inform how likely the hypothesis h is given data d, denoted by $P(h \mid d)$. The agent selects the hypothesis h that is most likely given data d.

To collect more data d, the Bayesian agent can perform different actions to interact with different sources of information. It chooses the action that yields the highest expected utility/value: it prefers to collect data from accurate sources of information (Griffiths et al., 2024). The agent determines the expected utility from an action a as follows:

$$EU(a) = \sum_{o \in \mathcal{O}_a} P(o \mid a, S)\, U(o).$$

Firstly, $\mathcal{O}_a$ is the set of all possible outcomes associated with an action a. The agent's belief state S summarizes the likelihood of observing each outcome o in $\mathcal{O}_a$. The agent uses this belief state S to compute the probability of observing an outcome o given a certain action a, denoted as $P(o \mid a, S)$. The agent multiplies this probability by the utility it would acquire from observing outcome o, denoted as $U(o)$. Then, the agent sums over all possible outcomes o in $\mathcal{O}_a$ ($\sum_{o \in \mathcal{O}_a}$) to determine the expected utility from action a.

To answer my research question, I used a classic ball-and-urn task as it is easy for children to understand, simple to model in a Bayesian manner, and well-established in the literature (Epstein and Halevy, 2019; Gelman et al., 2013; Kolossa et al., 2015; Kolossa, 2016; Xu and Garcia, 2008).Twelve-year-old children were faced with an urn containing white and black balls, with the true proportion of white balls (p_w) unknown to them; they

had to infer the proportion of white balls in the urn (p_{wa}). To achieve their objective, participants could draw one ball at a time from their own urn (personal sample) or ask for help from another "social" agent (social sample). Regarding the latter option, participants could obtain an estimate from a "social" agent who was estimating the proportion of white balls of a separate urn. They were also given the number of samples the "social" agent had drawn to produce that estimate. Given that this task requires a basic understanding of fractions, I restricted my participants to 12 years of age (Boyer et al., 2008). To elicit selective social learning, children were randomly assigned to one condition: in condition A, the social agent's urn was not identical to the participant's urn (personal urn: 30% white and social urn: 80% white); in condition B, the social agent's urn was identical (personal and social urn: 30% white). I predict that participants in condition A will draw more personal samples and less social samples as the social agent is inaccurate. Conversely, in condition B, I predict fewer personal samples and more social samples as the social agent is accurate. Furthermore, I posit that these behavioural patterns will be replicated by the Bayesian agent, suggesting that Bayesian agents can approximate selective social learning in children.

Methods

Participants

I randomly sampled 11 12-year-old children from a large junior high school institution in Ontario, Canada; 2 men and 9 women. The mean age of my subjects was 12.4 years with $SD = 0.5$ and range: 12.1-12.8.

Materials and Procedure

Participants completed my experiment on a "Streamlit" computer application. For the first step, my participants were presented with a prompt that explained what their task was and how they could accomplish this task (See Appendix A). They were told that they had to estimate the proportion of white balls of an urn that contained white and black balls. They could inform their estimate by drawing one ball at a time from their personal

urn, or by asking for the social agent's estimate of the proportion of white balls. Regarding the latter, participants were told that the social agent's urn might not be identical to their own and that it had accumulated 20 personal samples beforehand. Participants had 5 rounds to correctly guess the proportion of white balls. Within each round, participants could perform up to 15 personal and social samples. To mark the end of a round, the participant provided a proctor with their estimate who told them how close they were to the true proportion of white balls (1-7 Likert scale). For the second step, I taught my participants how discrete probability distributions work (See Appendix B). Then, my participants expressed their prior beliefs about their urn's true proportion of white balls: they were presented with a discrete number line [0,1] with 0.10 intervals; they distributed 30 chips across these intervals, where more chips in a given interval indicated a stronger belief that the true proportion of white balls fell within that range. Then, the participants completed the same task for the proportion of white balls that they thought would be stated by the social agent (See Appendix C). This information was needed for the Bayesian agent later on. For the last step, my participants were presented with the estimation task. Notably, I told them that they were sampling with replacement and that the proportion of white balls in the urn did not change across rounds. See Appendix D for what the task looked like.

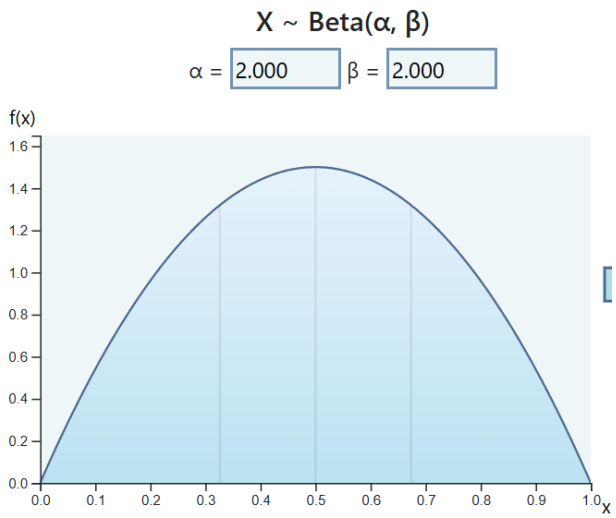
Bayesian Agent

The Bayesian agent was exposed to the same experimental paradigm as the participants. However, it only completed the last step under conditions A and B. Before interacting with the experiment, the Bayesian agent needed prior beliefs about the proportion of white balls in its own urn. It also required prior beliefs about the proportion of white balls reported by the social agent. These beliefs were expressed in terms of a Beta distribution with parameters α and β. To derive the value of these parameters, I used my participants' data from the second step (See Appendix F). On average, participants' average prior beliefs corresponded to $\alpha = 2$ and $\beta = 2$ in both cases. In words, participants

had a mild prior belief that the proportion of white balls in their own urn was around 0.5. A similar pattern was observed for the social agent's estimate.

Figure 1

Graphical Representation of Prior Beliefs With $\alpha = 2$ and $\beta = 2$



Note. The x-axis represents all possible proportions of white balls, ranging from 0 (all black) to 1 (all white). The y-axis is the probability density, which reflects the strength of the Bayesian agent's belief in each proportion: higher values indicate proportions the agent considers more plausible. In words, the symmetric peak at 0.5 reflects a prior belief that all proportions are plausible, with a stronger preference for proportions near 0.5.

Regarding personal sampling, p_{wa} is the agent's belief about the proportion of white balls in its own urn, which is approximated by the Beta(α, β) distribution. Roughly speaking, α and β represent the number of white and black balls, respectively. $\mathbb{E}[p_{wa}(\alpha, \beta)] = \frac{\alpha}{\alpha+\beta}$ is the Bayesian agent's estimate of the proportion of white balls from personal sampling. $\text{Var}[p_{wa}[\alpha, \beta)] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ reflects the uncertainty about said estimate. In terms of outcomes, the Bayesian agent observes a white or black ball being drawn. For an example, see Appendix G.

Concerning social sampling, p_{ws} is the agent's belief about the proportion of white

balls reported by the social agent; it is approximated by the Beta(α_s, β_s) distribution. Roughly speaking, α_s and β_s represent the number of white and black balls, respectively. $\mathbb{E}[p_{ws}(\alpha_s, \beta_s)] = \frac{\alpha_s}{\alpha_s+\beta_s}$ is the Bayesian agent's estimate of the proportion of white balls from social sampling. In terms of outcomes, the Bayesian agent observes the social agent's estimated proportion of white balls and the number of balls it has drawn so far. For an example, see Appendix H.

The Bayesian agent uses information from personal and social sampling to construct its overall belief state S. $\mathbb{E}[S]$ is the Bayesian agent's overall estimate of the proportion of white balls; this is the estimate that the proctor receives. S is a mixture distribution of p_{wa} and p_{ws} ($S = w_1 p_{wa} + w_2 p_{ws}$). In words, the Bayesian agent thinks that the true proportion of white balls in its urn may not be reflected in the social agent's estimate. As a result, the Bayesian agent assigns weight to the social sampling estimate $\mathbb{E}[p_{ws}]$ based on how close it is to the personal sampling estimate $\mathbb{E}[p_{wa}]$, scaled by its current uncertainty $\mathrm{Var}[p_{wa}]$ ($w_2 = e^{\frac{-|\mathbb{E}[p_{ws}]-\mathbb{E}[p_{wa}]|}{\mathrm{Var}[p_{wa}]}}$) If the social sampling estimate differs from the personal sampling estimate, then the Bayesian will not consider the social agent's estimate. This weighting mechanism gives rise to social selective learning.

In terms of utility, the Bayesian agent cares about reducing the uncertainty of its estimate $\mathrm{Var}[S]$, providing an accurate estimate $\mathbb{E}[S]$, and minimizing the cognitive cost from personal sampling. The Bayesian agent's utility depletes for each personal sample made (1). Prior research substantiates the inclusion of these three components: Individuals generally seek to minimize cognitive burden and uncertainty while maximizing accuracy (Hütter et al., 2016; Vasconcelos et al., 2022). Parameters in the utility function inform us how strongly the Bayesian agent wants to reduce uncertainty, provide an accurate estimate, and minimize cognitive cost (1).

When calculating the expected utility of personal sampling, the Bayesian agent relies on its current beliefs about the proportion of white balls in its own urn ($p_{wa}(\alpha, \beta)$) to estimate the chances of drawing a white or black ball. Specifically, the probability of

drawing a white ball is $\mathbb{E}[p_{wa}]$, and the probability of drawing a black ball is $1 - \mathbb{E}[p_{wa}]$. For each possible outcome, the agent calculates the utility of observing that outcome and multiplies it by the outcome's probability. Adding these weighted utilities together gives the expected utility of performing a personal sample (2).

Concerning the expected utility of social sampling, the agent uses its current beliefs about the social agent's proportion of white balls ($p_{ws}(\alpha_s, \beta_s)$) to construct a probability density function $f_{p_{ws}}$ over all possible proportions of white balls. This distribution roughly tells the agent how likely it would observe any proportion of white balls from social sampling. For each proportion, the Bayesian agent computes its utility and weights it by its probability under $f_{p_{ws}}$. Summing over all possible proportions gives the expected utility of social sampling (3).

The Bayesian agent chooses the action that yields the highest expected utility. If personal sampling and social sampling do not increase utility, then the agent will choose to state its estimate (4). Finally, if the Bayesian agent's estimate $\mathbb{E}[S]$ is from from the true porportion of white balls, then it will be more uncertain about its estimate (5). For more details on the mathematical structure of my Bayesian model, see Appendix I.

Design

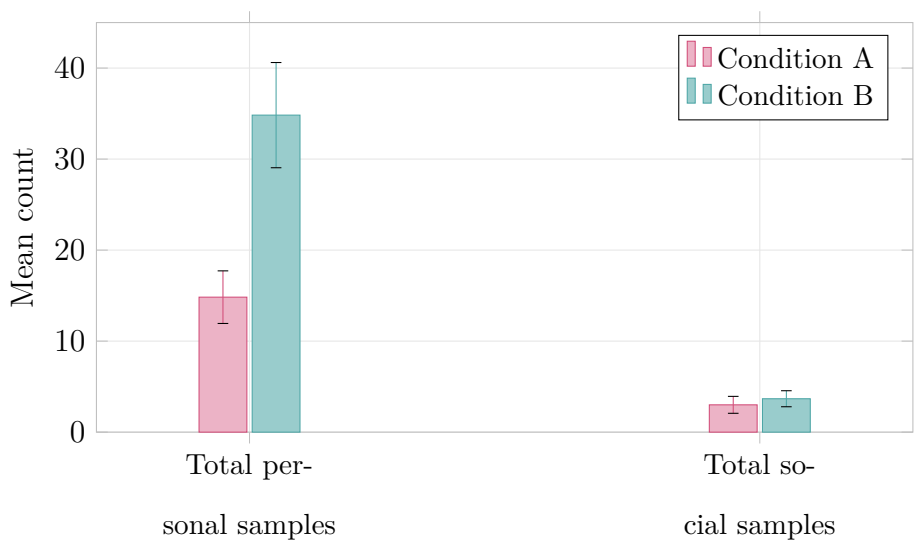
All participants were randomly assigned to condition A or condition B. For each condition, I tested whether the number of personal and social samples differed significantly between groups using a Welch's t-test as I did not want to assume equal variances across two small groups ($n_A = 6$, $n_B = 5$). To estimate the free parameters of the Bayesian model's utility function, I manually found the parameters that provided the best fit to my experimental data (1). I directly inputted the average prior beliefs $\alpha, \beta, \alpha_s, \beta_s$ across all participants into my Bayesian model. Then, the model completed conditons A and B, and I examined whether the Bayesian model's number of personal and social samples mirrored the patterns observed in the human data.

Results

Participants in condition A had a higher number of personal samples ($MD = 34.2$, $SD = 21.5$) than the ones in the condition B ($MD = 14.8$, $SD = 11.9$), $t = 1.80,\ \mathrm{df} = 6, p = .122,$ and $d = -1.15$. However, this difference was not statistically significant. In addition, the number of social samples did not significantly differ between condition A ($MD = 3.0$, $SD = 2.0$) and condition B ($MD = 4.2$, $SD = 5.6$), $t = -0.45,\ \mathrm{df} = 4.8, p = .672,$ and $d = -0.30$).

Table 1

Mean total personal and social samples for Conditions A and B across the five rounds



Note. Error bars show ± 1 *SE*. $n_A = 6$, $n_B = 5$.

Regarding my Bayesian model, the values of uncertainty aversion (θ), cognitive cost from personal sampling (τ), and responsiveness to error (γ) were 70, 0.25, and 1500, respectively (1). For condition A, my Bayesian model yielded 14 personal samples and 3 social samples whilst for condition B my Bayesian model completed 42 personal samples and 3 social samples.

Discussion

My Bayesian model converges well with the experimental data I obtained. This result suggests that children's selective learning is governed by the mechanisms present in the Bayesian model. To begin, children have the capability to cross-validate their estimate with the social agent's estimate by using a mixture distribution $S = w_1 p_{wa} + w_2 p_{ws}$ where $w_2 = e^{\frac{-|\mathbb{E}[p_{ws}]-\mathbb{E}[p_{wa}]|}{\mathrm{Var}[p_{wa}]}}$ and $w_1 + w_2 = 1$. In simpler terms, children cross-validate the social sampling estimate $\mathbb{E}[p_{ws}]$ with the personal sampling estimate $\mathbb{E}[p_{wa}]$ where the latter reflects their understanding of the world and the former represents information from external sources. If the social sampling estimate strongly deviates from their understanding of the world, then they will weigh the social agent's input less.

Congruent with my hypothesis, the number of personal samples between conditions A and B were very different. This behaviour can be explained by my Bayesian model: Children care about minimizing the uncertainty of their estimate $\mathrm{Var}[S]$ and the cognitive cost from personal sampling n_{ps} as mentioned in the utility function (1). In condition B, w_2 was approximately 0 as the social agent's estimate did not reflect the proportion of white balls in the personal urn. Therefore, $w_1 \approx 1$ in condition B so the mixture's variance $\mathrm{Var}[S(p]$ was driven almost entirely by the personal sampling distribution $p_{wa}(\alpha, \beta)$. Drawing a personal sample dramatically reduces $\mathrm{Var}[S]$, yielding a high utility per personal sample. In condition A, w_2 and w_1 were similar as the social agent's estimate approximated the true proportion of white balls in the personal urn. Thus, $w_1 \approx 0.5$, the personal sampling distribution contributes equally to the mixture variance, so taking a personal sample did not reduce $\mathrm{Var}[S]$ as much as in condition B, yielding a lower utility per personal sample.

Contrary to my hypothesis, the number of social samples did not differ substantially between conditions A and B. However, this finding is also consistent with my Bayesian model: the estimated cognitive cost of personal sampling ($\tau = 0.25$) was relatively low, meaning that participants did not expend much cognitive effort when estimating the

proportion of white balls themselves. Since personal sampling reduced variance $\mathrm{Var}[S]$ and was cognitively inexpensive, participants had little incentive to seek social information, which likely accounts for the low social sampling rates across conditions.

Regarding limitations and future directions, the current model was only evaluated against the total number of personal and social samples aggregated across all rounds. Notably, I did not assess whether the model's predictions mapped onto sampling behaviour at the round level, nor whether it correctly predicted individual participants' actions at each timestep. Future work should examine both of these questions.

My research has important implications for developmental psychology. My results suggest I was able to uncover the main mechanism that governs selective learning. In addition, my results suggest that children reason in a Bayesian manner under selective social learning tasks. Taken together, these suggestions provide cognitive scientists with strong evidence that computational models of the human behaviour should be Bayesian. Moreover, via simulation, researchers can use my Bayesian model to answer a variety of causal effect questions. For example, they could directly manipulate uncertainty aversion and obtain results that would reflect a child's behaviour.

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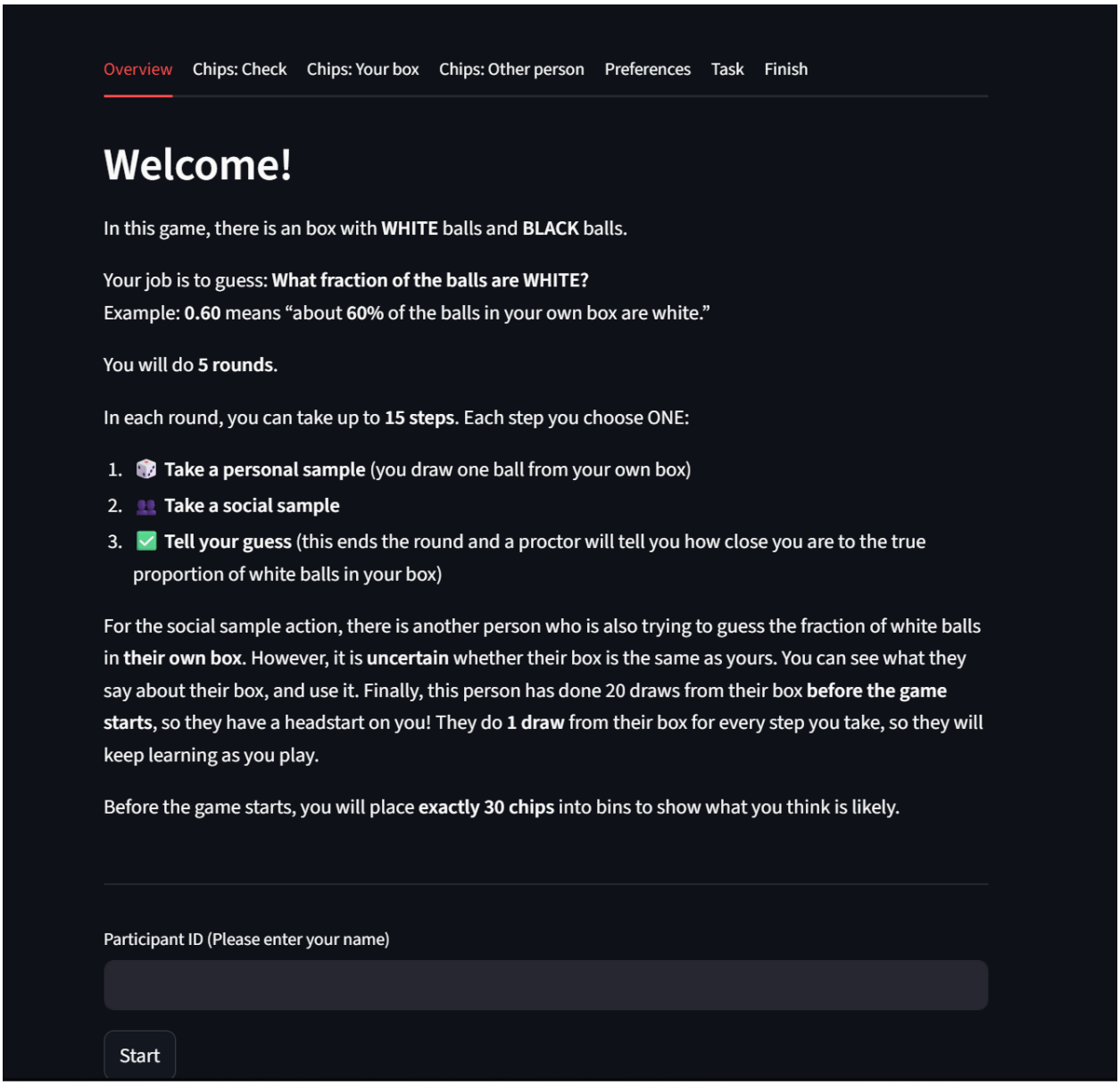
Appendix A

Introduction Screen



Appendix B

Comprehension Checks

Figure B1

Comprehension Check 1

Comprehension Check Before the Chips Task

The numbers below represent possible proportions of white balls in the box. Place more chips on the proportions you think are more likely. Place fewer chips on the proportions you think are less likely. Here are two examples to show what chip placement can look like.

✅ Example 1

Suppose you think that:

- **0.5 to 0.8 are equally likely, meaning that a 50% to 80% proportion of white balls in the box are equally likely.**
- **0.0 to 0.1 is less likely, meaning a 0% to 10% proportion of white balls in the box are less likely**

With **30 chips**, one possible placement could be:

	Bin	Chips
0	0.0–0.1	1
1	0.1–0.2	2
2	0.2–0.3	2
3	0.3–0.4	3
4	0.4–0.5	4
5	0.5–0.6	6
6	0.6–0.7	6
7	0.7–0.8	6
8	0.8–0.9	0
9	0.9–1.0	0

Figure B2

Comprehension Check 2

✅ Example 2

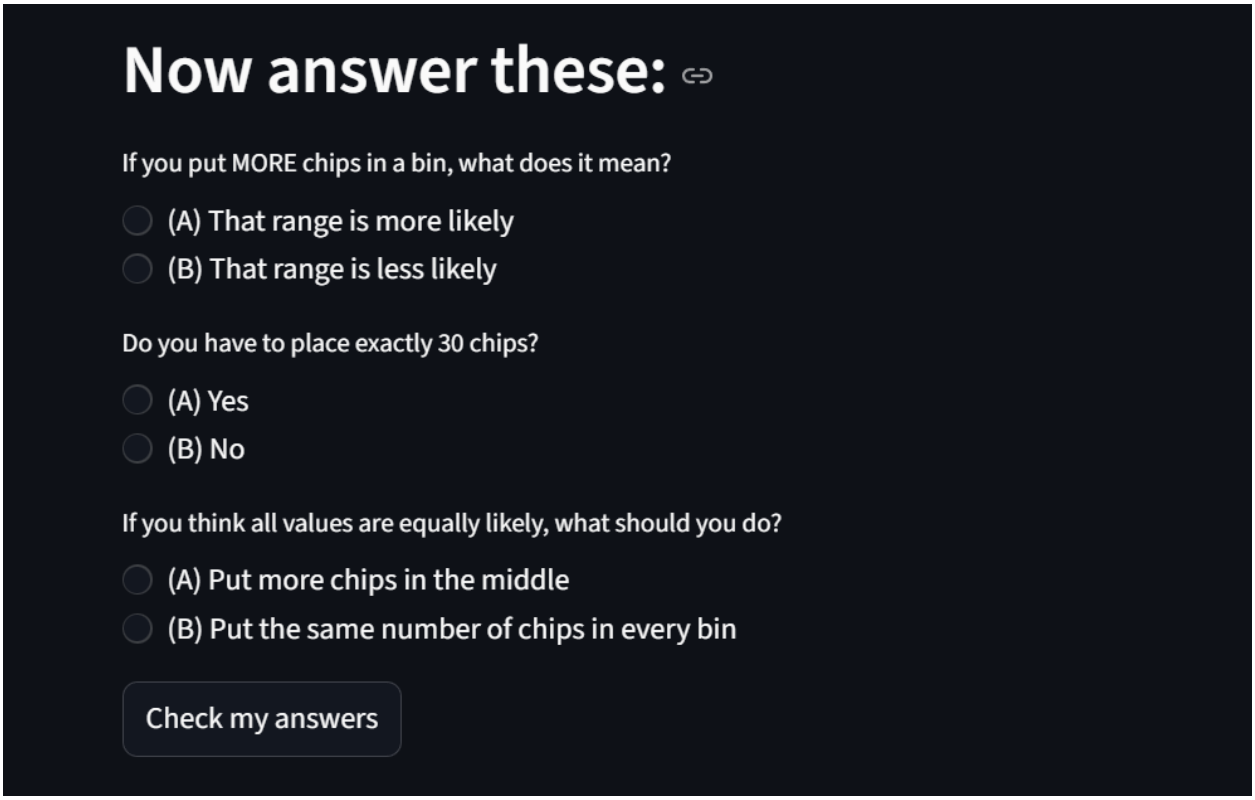
Suppose you think **all fractions are equally likely.**

With **30 chips** and 10 bins, that means **3 chips in every bin**:

	Bin	Chips
0	0.0–0.1	3
1	0.1–0.2	3
2	0.2–0.3	3
3	0.3–0.4	3
4	0.4–0.5	3
5	0.5–0.6	3
6	0.6–0.7	3
7	0.7–0.8	3
8	0.8–0.9	3
9	0.9–1.0	3

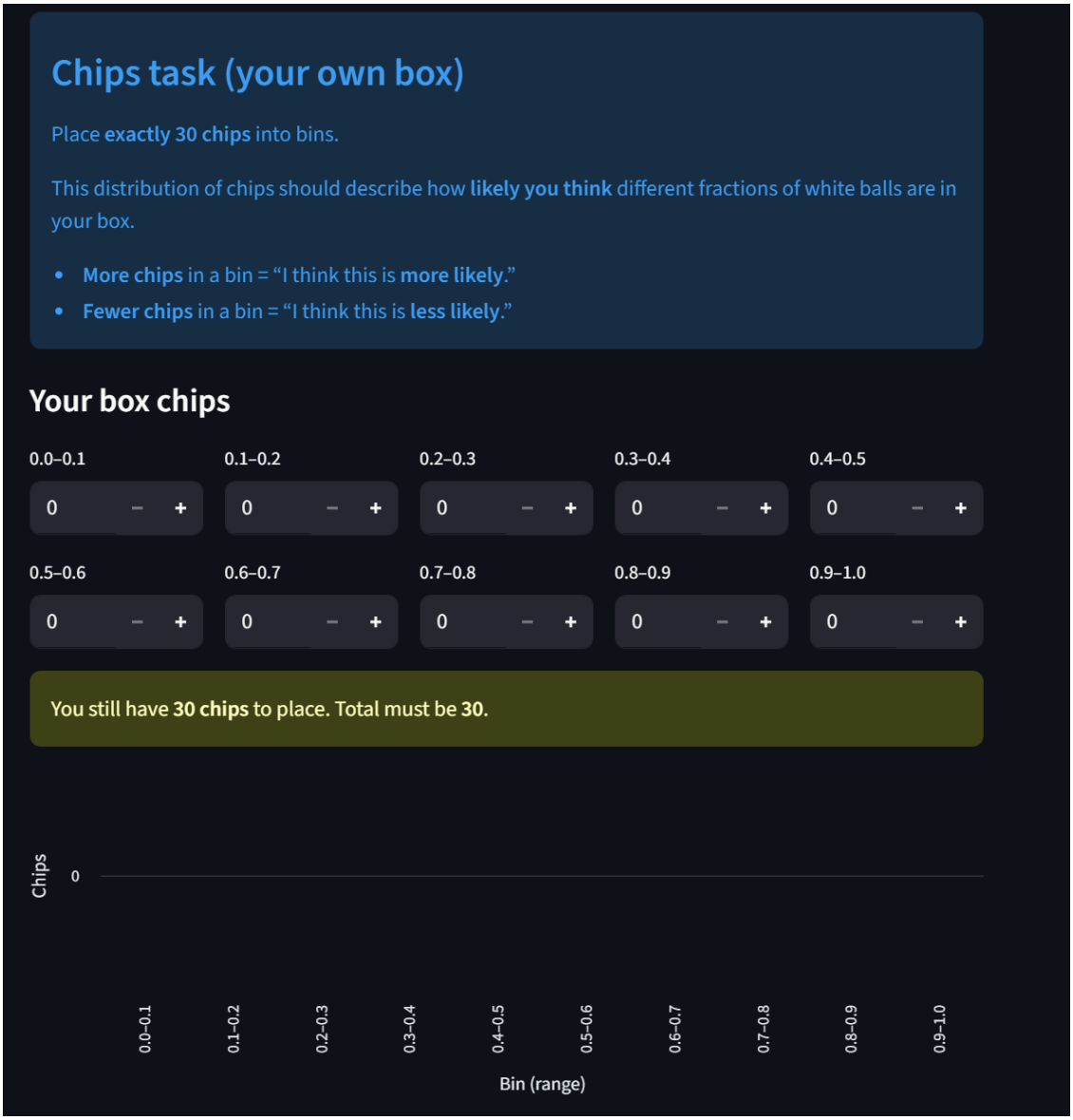
Figure B3

Comprehension Check 3



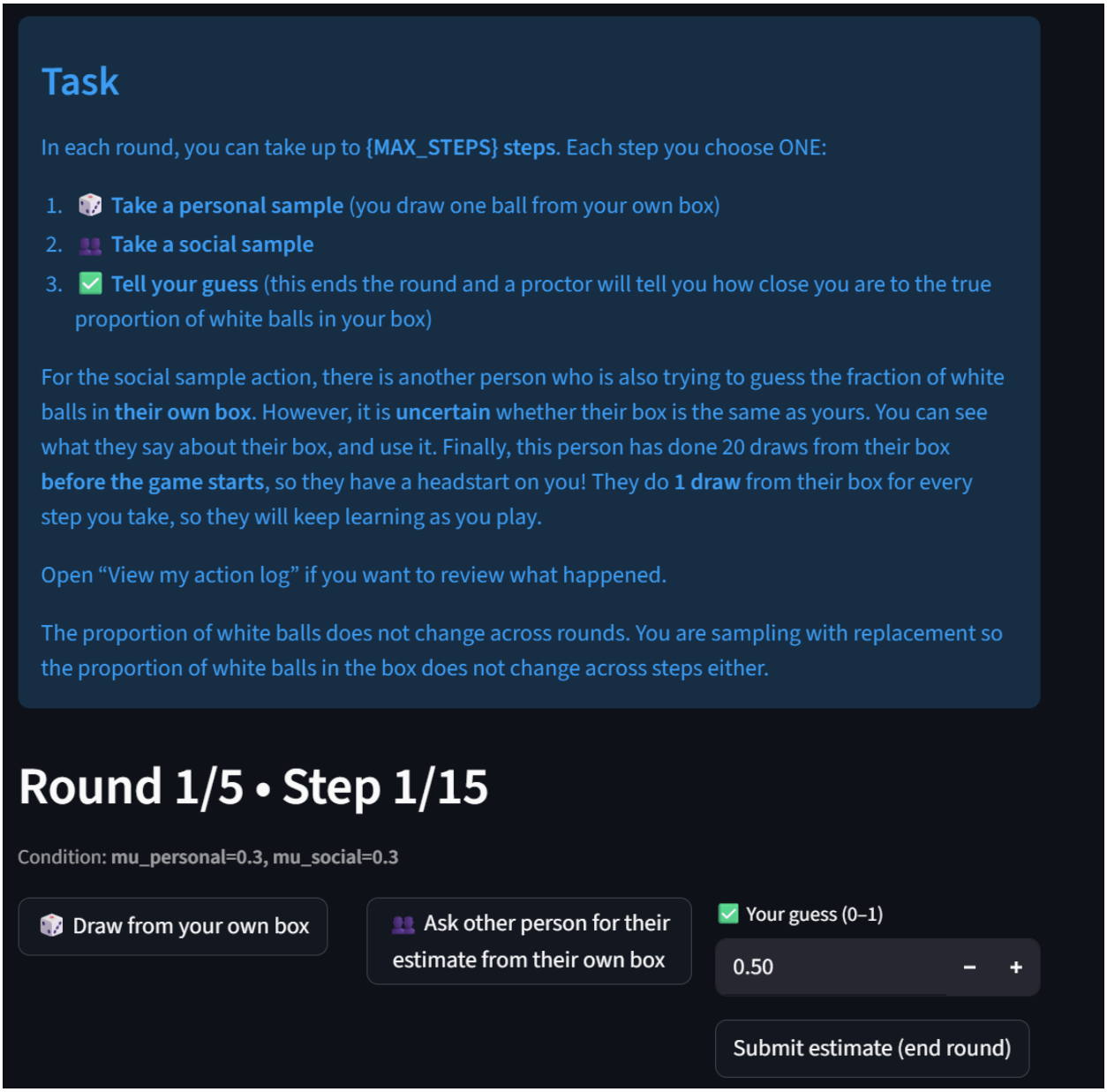
Appendix C

Elicitation of Prior Beliefs About the Distribution of White Balls in Your Own Urn



Appendix D

Task Screen



Appendix E

Table 1

Welch's Independent-Samples t-Tests Comparing Model Parameters Between Conditions A and B

Parameter	**Mean A**	**Mean B**	t	df	p	d	**sig**
Uncertainty Aversion (θ)	7.583	8.100	−0.384	6.4	.713	−0.215	ns
Responsiveness to Error (γ)	7.167	7.900	−0.646	7.3	.538	−0.366	ns
Cognitive cost (τ)	4.250	3.800	0.292	9.0	.777	0.173	ns
$\alpha_{0,p}$	2.755	1.205	1.233	5.3	.270	0.679	ns
$\beta_{0,p}$	2.877	1.205	1.046	5.2	.342	0.575	ns
$\alpha_{0,s}$	3.952	2.251	0.831	6.3	.436	0.465	ns
$\beta_{0,s}$	3.381	2.056	0.668	6.4	.528	0.374	ns

$n_A = 6$, $n_B = 5$. d = Cohen's d effect size. Significance codes: $^{***}p < .001$, $^{**}p < .01$, $^{*}p < .05$, . $p < .10$, ns $p \geq .10$.

Appendix F

Fitting Discrete Probability Distributions

Behind the scenes, I fit participants' discrete probability distributions to a Beta distribution, which is commonly used to model prior beliefs about binary outcome distributions (See Bayesian Model section). It uses method of moments estimates as a starting point $(\hat{\alpha}_0, \hat{\beta}_0)$, then minimizes the negative log-likelihood over the number of chips within each interval, where each interval's probability is the Beta cumulative density function mass over that interval: $p_i(\alpha, \beta) = F_{\text{Beta}}(e_{i+1};\, \alpha, \beta) - F_{\text{Beta}}(e_i;\, \alpha, \beta)$
$(\hat{\alpha},\, \hat{\beta}) = \arg\min_{\alpha,\beta>0} \left[-\sum_{i=1}^{K} c_i \log p_i(\alpha, \beta) \right].$

Appendix G

Personal Sampling Update Example

For example, suppose that the Bayesian agent holds a prior belief $p_{wa}(2,2)$ before sampling from the urn; its current belief is $p_{wa}(2,2)$ with expectation $\mathbb{E}[p_{wa}(2,2)] = \frac{2}{4} = 0.5$, and variance $\mathrm{Var}[p_{wa}(2,2)] = \frac{2 \cdot 2}{(4)^2(5)} = \frac{4}{80} = 0.05$. If the agent draws a black ball, the parameters are updated to $\alpha = 2 + 0 = 2$ and $\beta = 2 + 1 = 3$. The agent's updated belief is therefore $p_{wa}(2,3)$, with expectation $\mathbb{E}[p_{wa}(2,3)] = \frac{2}{5} = 0.4$, and variance $\mathrm{Var}[p_{wa}(2,3)] = \frac{2 \cdot 3}{(5)^2(6)} = \frac{6}{150} = 0.04$.

Appendix H

Social Sampling Update Example

For example, suppose that the Bayesian agent assumes a prior of $\alpha_s = 2$ and $\beta_s = 2$, corresponding to an expected proportion of $\mathbb{E}[p_{ws}(2,2)] = \frac{2}{4} = 0.5$ and a variance of $\mathrm{Var}[p_{ws}(2,2)] = \frac{2 \cdot 2}{(4)^2(5)} = \frac{4}{80} = 0.05$. If the social agent reports that it has drawn 100 balls and that its expected proportion is 0.3, then the Bayesian agent updates its beliefs as follows: $\alpha_s = 2 + (0.3)(100) = 32$ and $\beta_s = 2 + (1 - 0.3)(100) = 72$, yielding an expected proportion of $\mathbb{E}[p_{ws}(32,72)] = \frac{32}{32+72} = \frac{32}{104} = 0.31$ and a variance of $\mathrm{Var}[p_{ws}(32,72)] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{32 \cdot 72}{(32+72)^2(32+72+1)} = \frac{32 \cdot 72}{(104)^2(105)} = \frac{2304}{1{,}135{,}680} \approx 0.00203$.

Appendix I

Bayesian Model

Let agent 1 be the Bayesian agent. Let agent 2 be the social agent. Let p_w be the true proportion of white balls in agent 1's urn. Let $X \in \{1, 0\}$ represent the color of a ball randomly drawn from a Bernoulli distribution, where $X = 1$ indicates white and $X = 0$ indicates black, with $\Pr(X = 1) = p$ and $\Pr(X = 0) = 1 - p$. The ball distribution in the urn is approximated by a Bernoulli distribution with $\mathbb{E}[X] = p$ and $\mathrm{Var}[X] = p(1 - p)$.

Let p_{wa} denote agent 1's belief about the proportion of white balls in its own urn, where $p_{wa} \sim \mathrm{Beta}(\alpha, \beta)$, $\mathbb{E}[p_{wa}(\alpha, \beta)] = \frac{\alpha}{\alpha+\beta}$, and $\mathrm{Var}[p_{wa}(\alpha, \beta)] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$. To refine this belief, agent 1 can draw a ball from its own urn with replacement (personal sampling).

Let p_{ws} denote agent 1's belief about the proportion of white balls reported by agent 2, where $p_{ws} \sim \mathrm{Beta}(\alpha_s, \beta_s)$, $\mathbb{E}[p_{ws}(\alpha_s, \beta_s)] = \frac{\alpha_s}{\alpha_s+\beta_s}$, and $\mathrm{Var}[p_{ws}(\alpha_s, \beta_s)] = \frac{\alpha_s\beta_s}{(\alpha_s+\beta_s)^2(\alpha_s+\beta_s+1)}$. To refine its belief, agent 1 can solicit an estimate from agent 2: agent 2 provides agent 1 with an estimated proportion of white balls along with its current sample size, drawn from an urn with a potentially different composition (social sampling).

Agent 1's belief state is defined as $S = w_1 p_{wa} + w_2 p_{ws}$, a mixture distribution with weights $w_1, w_2 \in \mathbb{R}, w_1 + w_2 = 1$. The expectation of this belief state S represents agent 1's estimate of the proportion of white balls in its urn. Agent 1 is skeptical about whether agent 2 observed the same urn. As a result, agent 1 weights the social estimate $\mathbb{E}[p_{ws}]$ in proportion to its proximity to $\mathbb{E}[p_{wa}]$, scaled by agent 1's current uncertainty of its personal urn estimate $\mathrm{Var}[p_{wa}]$. The social weight is $w_2 = e^{\frac{-|\mathbb{E}[p_{ws}]-\mathbb{E}[p_{wa}]|}{\mathrm{Var}[p_{wa}]}}$. In words, if the estimate from social sampling strongly differs from the estimate from personal sampling, then the Bayesian agent will not consider the agent 2's estimate. This weighting mechanism gives rise to social selective learning within agent 1.

The experiment consists of five rounds r, each containing 15 time steps t. As a result, all variables have the subscripts t and r. Within each round, agent 1 selects from

the action set $A = \{a_{ps}, a_{ss}, a_{sp}\}$, where a_{ps} denotes personal sampling, a_{ss} denotes social sampling, and a_{sp} denotes stating an estimate. Each action $a \in A$ yields an observation from a corresponding outcome set: $\mathcal{O}_{ps} = \{\text{white}, \text{black}\}$, $\mathcal{O}_{ss} = \{p_{ws} : j \in \text{Social sample}\}$, and $\mathcal{O}_{sp} = \{\varnothing\}$. At each step t within round r, agent 1 selects action $a_{t,r}$ given state $S_{t,r}$ by computing the distribution over possible outcomes $o_{a,t,r} \in \mathcal{O}_a$ and choosing the action with the highest expected utility EU. After observing outcome $o_{t,r}$, the agent updates its belief state via $S_{t+1,r}(o_{t,r}) = f(S_{t,r}, a_{t,r}, o_{t,r})$.

$U(S, t, r) = -\theta\sqrt{\text{Var}[S_{t,r}]} - \gamma(|p_w - \mathbb{E}[S_{t,r}]|) - \tau(n_{ps,t,r})$ (1) is agent 1's utility/reward function with $\theta, \gamma, \tau \in \mathbb{R}^+$ where θ, γ, τ is agent 1's uncertainty aversion, responsiveness to error, and cognitive cost associated with personal sampling, respectively. Additionally, $n_{ps,t,r}$ is the cumulative number of personal samples taken at time t in round r. Notably, $\gamma = 0$ within each round, since the proctor only discloses how close agent 1's estimate was to p_w at the end of the round.

Agent 1 chooses the action at time t and round r that yields the highest expected reward (2):

$$a^*_{t,r} = \begin{cases} a_{sp,t,r}, & \text{if } U(S_{t,r}) > \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ps,t,r},\ S_{t,r}] \\ & \quad \wedge\ U(S_{t,r}) > \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ss,t,r},\ S_{t,r}] \\ a_{ps,t,r}, & \text{if } U(S_{t,r}) \leq \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ps,t,r},\ S_{t,r}] \\ & \quad \wedge\ \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ps,t,r},\ S_{t,r}] > \mathbb{E}[U(S_{t+1,r}) \mid a_{t,r} = a_{ss,t,r},\ S_{t,r}] \\ a_{ss,t,r}, & \text{if } U(S_{t,r}) \leq \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ss,t,r},\ S_{t,r}] \\ & \quad \wedge\ \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ss,t,r},\ S_{t,r}] \geq \mathbb{E}[U(S_{t+1,r}) \mid a_{t,r} = a_{ps,t,r},\ S_{t,r}] \end{cases}$$

Agent 1's expected utility under social sampling is (3):

$$
\begin{aligned}
\mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ss,t,r}, S_{t,r}] &= \int_0^1 U(S_{t+1,r}(o_{t,r}))\, f_{p_{ws,t,r}}(o_{t,r})\, do_{t,r} \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} U(S_{t+1,r}(o_{i,t,r}))\, f_{p_{ws,t,r}}(o_{i,t,r})\, \Delta o \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} \Bigg(-\theta\sqrt{\mathrm{Var}[S_{t+1,r}(o_{i,t,r})]} \\
&\qquad - \gamma \Big| p_w - \mathbb{E}[S_{t+1,r}(o_{i,t,r})] \Big| - \tau(n_{ps,t,r}) \Bigg) f_{p_{ws,t,r}}(o_{i,t,r})\, \Delta o \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} \Bigg(-\theta\sqrt{\mathrm{Var}[w_1 p_{wa,t,r}(\alpha_{t,r}, \beta_{t,r}) + w_2 p_{ws,t,r}(\alpha_{s,t+1,r}, \beta_{s,t+1,r})]} \\
&\qquad - \gamma \Big| p_w - \mathbb{E}[w_1 p_{wa,t,r}(\alpha_{t,r}, \beta_{t,r}) + w_2 p_{ws,t,r}(\alpha_{s,t+1,r}, \beta_{s,t+1,r})] \Big| \\
&\qquad - \tau(n_{ps,t+1,r}) \Bigg) f_{p_{ws,t,r}}(o_{i,t,r})\, \Delta o \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} \Bigg(-\theta\sqrt{\mathrm{Var}[w_1 p_{wa,t,r}(\alpha_{t,r}, \beta_{t,r})} \\
&\overline{+ w_2 p_{ws,t,r}(\alpha_{s,t,r} + (k_{s,t,r}) o_{i,t,r}, \beta_{s,t,r} + (k_{s,t,r})\,(1 - o_{i,t,r}))]} \\
&\qquad - \gamma \Big| p_w - \mathbb{E}[w_1 p_{wa,t,r}(\alpha_{t,r}, \beta_{t,r}) + \\
&\quad w_2 p_{ws,t,r}(\alpha_{s,t,r} + (k_{s,t,r}) o_{i,t,r}, \beta_{s,t,r} + (k_{s,t,r})\,(1 - o_{i,t,r}))] \Big| \\
&\qquad - \tau(n_{ps,t,r}) \Bigg) f_{p_{ws,t,r}}(o_{i,t,r})\, \Delta o
\end{aligned}
$$

where $o_{t+1,r} \in \mathcal{O}_{ss}$, $\Delta o = \frac{1-0}{n}$, $o_{i,t+1,r} = 0 + i\,\Delta o$, and $f_{p_{ws,t,r}}(o_{t,r}) = \frac{1}{B(\alpha_{s,t,r}, \beta_{s,t,r})} o_{t,r}^{\alpha_{s,t,r}-1} (1 - o_{t,r})^{\beta_{s,t,r}-1}$.

Upon receiving agent 2's estimate, agent 1 revises its beliefs about agent 2's

proportion of white balls as follows:

$$\alpha_{s,t+1,r} = \alpha_{s,t,r} + (k_{s,t,r})\hat{p}_{ws,t,r}$$

$$\beta_{s,t+1,r} = \beta_{s,t,r} + (k_{s,t,r})\,(1 - \hat{p}_{ws,t,r})$$

where $k_{s,t,r}$ and $\hat{p}_{ws,t,r}$ are the sample size and white-ball proportion reported by agent 2 at time t, and $\alpha_{s,t,r}$, $\beta_{s,t,r}$ encode agent 1's prior beliefs about agent 2's estimate.

Regarding personal sampling, agent 1's expected utility is defined as follows (4):

$$\begin{aligned}
\mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ps,t,r}, p_{wa,t,r}] &= \sum_{o_{t,r} \in \mathcal{O}_{ps}} U(S_{t+1,r}(o_{t,r}))\, P(o_{t,r} \mid p_{wa,t,r}, a_{ps,t,r}) \\
&= P(X_{t,r} = 1 \mid p_{wa,t,r})\, U(S_{t+1,r}(X_{t,r} = 1)) \\
&\quad + \Big(1 - P(X_{t,r} = 1 \mid p_{wa,t,r})\Big)\, U(S_{t+1,r}(X_{t,r} = 0)) \\
&= \mathbb{E}[p_{wa,t,r}]\, U\Big(w_1\, p_{wa,t+1,r}(\alpha_{t,r} + 1, \beta_{t,r}) + w_2\, p_{ws,t,r}\Big) \\
&\quad + \Big(1 - \mathbb{E}[p_{wa,t,r}]\Big)\, U\Big(w_1\, p_{wa,t+1,r}(\alpha_{t,r}, \beta_{t,r} + 1) + w_2\, p_{ws,t,r}\Big) \\
&= \mathbb{E}[p_{wa,t,r}]\Bigg[- \theta\sqrt{\mathrm{Var}(p_{wa,t+1,r}(\alpha_{t,r} + 1, \beta_{t,r}))} \\
&\qquad - \gamma\Big|p_w - \mathbb{E}[p_{wa,t+1,r}(\alpha_{t,r} + 1, \beta_{t,r})]\Big| - \tau(n_{ps,t,r} + 1)\Bigg] \\
&\quad + \Big(1 - \mathbb{E}[p_{wa,t,r}]\Big)\Bigg[- \theta\sqrt{\mathrm{Var}(p_{wa,t+1,r}(\alpha_{t,r}, \beta_{t,r} + 1))} \\
&\qquad - \gamma\Big|p_w - \mathbb{E}[p_{wa,t+1,r}(\alpha_{t,r}, \beta_{t,r} + 1)]\Big| - \tau(n_{ps,t,r} + 1)\Bigg].
\end{aligned}$$

When agent 1 chooses to report its estimate (i.e., takes action a_{sp}), it states its current expected belief $\mathbb{E}[S_{t,r}]$, after which a proctor provides feedback indicating how close this estimate is to the true proportion p_w. The agent then proceeds to the next round, carrying forward both the cumulative number of personal samples n_{ps} from previous rounds

and its final belief from the most recent round, denoted $\mu_{m,n} = \mathbb{E}[S_{m,n}]$. To account for estimation error, an uncertainty penalty is applied: if the agent's estimate is far from the true proportion, it becomes less confident in its beliefs. This is implemented by increasing the variance of the agent's Beta distribution over p_{wa} while keeping its mean unchanged. Formally, the penalty is defined as $p_{u,m,n} = \lambda\Big(|p_w - p_{wa,m,n}|\Big)$, where $\lambda \in \mathbb{R}^+$, and the agent's prior confidence is captured by the precision parameter $\kappa_{m,n} = \alpha_{m,n} + \beta_{m,n}$. To increase uncertainty, this precision is rescaled as $\kappa_{0,n+1} = \kappa_{m,n}/p_{u,m,n}$, and the updated parameters are given by $\alpha_{0,n+1} = \mu_{m,n}\kappa_{0,n+1}$ and $\beta_{0,n+1} = (1 - \mu_{m,n})\kappa_{0,n+1}$. This transformation increases the variance of the Beta distribution without altering its expected value. (5)